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Trends and Methodological Advances in Modeling Heavy Metal Exposure During Pregnancy: A Combined Bibliometric and Systematic Review

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Abstract: Exposure to heavy metals during pregnancy poses significant risks to maternal and child health. This study aims to analyze trends in research on heavy metal exposure in pregnant women using bibliometric analysis and systematic review. A total of 125 publications from the Scopus database (1973–2023) were analyzed. The bibliometric analysis examined publication trends, geographical distribution, and researcher collaboration. A systematic review assessed statistical methods, focusing on advanced models like Bayesian Kernel Machine Regression (BKMR) and Weighted Quantile Sum (WQS) regression, as well as traditional methods like linear and logistic regression. The bibliometric analysis showed an increasing number of studies, with the United States leading in publications, followed by China and India. The systematic review found that advanced models such as BKMR and WQS regression are increasingly used to evaluate interactions between heavy metal mixtures and pregnancy outcomes. However, many studies still rely on traditional regression models, highlighting a gap in the adoption of more advanced techniques. The study highlights the particular analytical capabilities of advanced statistical models in examining complex interactions between heavy metal mixtures and pregnancy outcome in understanding complex interactions between heavy metal mixtures and pregnancy outcomes. It also points to the need for greater adoption of these methods to improve risk assessments and inform public health policies. In conclusion, advanced modeling techniques, particularly BKMR and WQS regression, are crucial for understanding the risks of heavy metal exposure during pregnancy and guiding public health interventions.

Keywords: Pregnancy; Heavy metals; Bibliometric analysis; Health policy

1 Introduction

Adverse pregnancy and birth outcomes, such as miscarriage, low birth weight (LBW), preterm birth (PTB), and small for gestational age (SGA), remain persistent public health concerns worldwide due to their association with increased maternal and neonatal morbidity and mortality [1–3]. These outcomes can lead to long-term developmental consequences, including cognitive impairment, growth retardation, and an elevated risk of chronic diseases such as cardiovascular disorders, obesity, hypertension, and type 2 diabetes in later life [4–7].

Environmental exposures, particularly to heavy metals during pregnancy, are increasingly recognized as major risk factors influencing these adverse outcomes [8, 9]. Toxic metals such as lead (Pb), cadmium (Cd), and mercury

(Hg) have been linked to reproductive complications and fetal developmental delays, including reduced birth weight, shorter body length, decreased head and chest circumference, and impaired neurodevelopment [9]. These metals are commonly found in industrial, agricultural, and high-traffic environments, with exposure levels often exceeding safe thresholds, particularly in low- and middle-income countries [10–15]. According to the World Health Organization (WHO), millions of people, especially pregnant women, are exposed to hazardous levels of heavy metals annually, leading to negative health outcomes in both adults and children [10, 16]. For example, a study in Kao Bay, Indonesia, found a risk quotient exceeding 1 from fish consumption, indicating that it posed a potential health risk for human consumption [16].

Birth weight is an important pregnancy outcome that can be influenced by multiple maternal, fetal, and environmental factors, including prenatal exposure to heavy metals, whereas maternal or cord-blood metal concentrations provide more direct biomarkers of exposure [17, 18]. Previous studies have shown significant associations between prenatal exposure to metals such as Cadmium (Cd) and Chromium (Cr) and negative birth outcomes, including LBW and smaller head circumference [11, 19]. Additionally, statistical modeling has linked multi-metal exposure to an increased risk of PTB [20].

Despite increasing awareness, research examining the interactions of metal mixtures and their combined effects on maternal and neonatal outcomes remains limited. Some studies have demonstrated associations between prenatal exposure to metals and congenital anomalies, such as congenital heart defects, with lead being identified as a major contributor using Bayesian Kernel Machine Regression (BKMR) [21–23]. However, traditional statistical models like linear and logistic regression have shown limitations in capturing complex, nonlinear interactions among co-exposures [24].

Given the growing body of evidence and the complexity of metal mixtures, there is a critical need to assess the methodological rigor of existing studies. Currently, no comprehensive systematic review utilizing bibliometric analysis exists to evaluate the modeling approaches used to investigate heavy metal exposure during pregnancy and its impact on birth outcomes.

Traditional regression models, while widely used, treat each exposure in isolation and assume linearity, potentially underestimating true risk relationships. They lack sensitivity to complex mixture effects and nonlinear dynamics. In contrast, advanced models such as BKMR and Weighted Quantile Sum (WQS) regression have emerged as more powerful tools for analyzing environmental mixtures. These approaches provide different strategies for characterizing mixture effects: WQS estimates a directional weighted index under specified assumptions, whereas BKMR provides a more flexible kernel-based framework for modeling potentially nonlinear and interactive exposure–response relationships. However, the adoption of these advanced techniques remains limited, and many studies still rely on conventional approaches. Furthermore, existing research often lacks rigorous sensitivity analyses, fails to account for key confounders (e.g., maternal nutrition, socioeconomic status), or does not validate model predictions.

This study aims to bridge this gap by conducting a bibliometric analysis and systematic review of the current literature to examine global research trends in prenatal heavy metal exposure modeling, identify the most frequently used statistical models, and evaluate their respective strengths, limitations, and effectiveness in assessing the impact of heavy metal exposure on pregnancy outcomes. This review further contributes to Sustainable Development Goal 3, which advocates for improved maternal and child health and reduced mortality and morbidity associated with exposure to hazardous chemicals [25]. The novelty of this manuscript lies in its dual analytical framework integrating bibliometric analysis and systematic review—to not only identify key research trends but also critically compare the performance of advanced statistical models such as BKMR and WQS regression.

2 Methodology

2.1 Study Design

This study employed a two-pronged approach combining bibliometric analysis and systematic review to explore global research trends and evaluate statistical modeling techniques applied to studies of heavy metal exposure among pregnant women. The systematic review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement [26].

2.2 Bibliometric Analysis

2.2.1 Data source and search strategy

Bibliometric data were exclusively sourced from the Scopus database, chosen for its extensive coverage of peer-reviewed scientific literature and consistent indexing of citation and author affiliation metadata. The search strategy employed a combination of the keywords “pregnant women,” “heavy metals,” and “modeling” The literature search was conducted in Scopus using the TITLE-ABS-KEY fields and a predefined Boolean search strategy. The exact search query, search date, publication cutoff, and selection criteria for the 125 records included in the bibliometric analysis are provided in the Supplementary Material. Publications from the period 1973 to 2023 were retrieved. Initially, 1,564 records were identified from Scopus. Following document type filtering and relevance screening,

125 publications pertinent to the statistical modeling of heavy metal exposure during pregnancy were retained for bibliometric analysis. After a full-text eligibility assessment, 11 studies met the inclusion criteria for the systematic review. To ensure consistency in citation analysis, co-occurrence mapping, and bibliometric visualization, all analyses were conducted exclusively using Scopus-indexed records.

2.2.2 Screening and selection

The screening and selection process adhered to the PRISMA 2020 guidelines (Figure 1). The Scopus search (up to 31 December 2023) initially retrieved 1,564 records. Restricting the results to journal articles removed 782 records of other document types [specify, e.g., conference papers, reviews, book chapters], leaving 782 journal articles for screening. Of these, 715 records were excluded: 21 non-English articles, 56 duplicate articles, and 638 records based on title and abstract screening. The remaining 67 articles underwent full-text assessment, of which 56 were excluded because their results were irrelevant to the study objectives, resulting in 11 studies included in the systematic review. Separately, the bibliometric analysis was conducted on 125 documents [state the selection criteria, e.g., drawn from the 782 screened articles], which is a broader set than the 67 full-text articles and therefore not a subset of the systematic review sample.

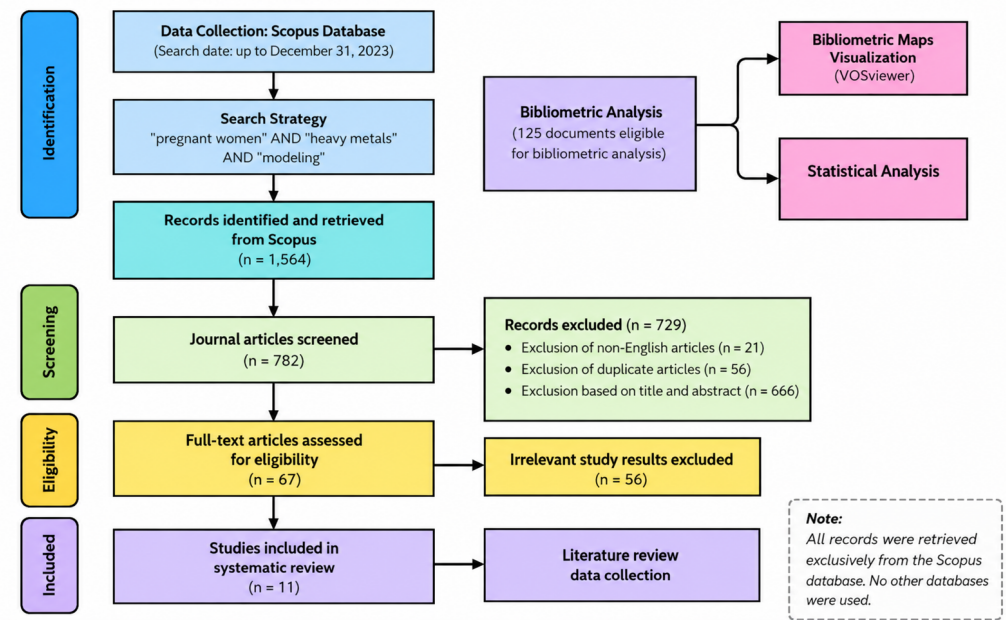


Figure 1. PRISMA flow diagram illustrating the identification, screening, eligibility assessment, and inclusion of studies retrieved exclusively from the Scopus database

Note: PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

2.2.3 Data cleaning and analysis

The dataset was curated using OpenRefine for normalization of author names and metadata. Cleaned data were organized in Microsoft Excel. Bibliometric indicators such as publication year, country of origin, and citation count were analyzed. VOSviewer software 1.6.20 was employed to generate visual maps illustrating author collaboration networks and keyword co-occurrence patterns.

2.3 Systematic Review

2.3.1 Inclusion and exclusion criteria

Eligible studies met the following criteria:

Studies were eligible if they were peer-reviewed, published in English, involved pregnant women exposed to at least one heavy metal, and used quantitative modeling (e.g., regression, BKMR, or WQS). Eligible outcomes included birth outcomes (e.g., birth weight, gestational age, or congenital anomalies) and postnatal neurobehavioral outcomes assessed from birth to 5 years of age. Studies were excluded if they lacked a modeling approach, did not involve pregnancy, or did not report eligible birth or early-life outcomes.

2.3.2 Data extraction

Two independent reviewers conducted data extraction using a standardized form. Extracted information included study design, sample size, exposure and outcome variables, modeling technique, and key findings. Discrepancies

were resolved by consensus or adjudicated by a third reviewer. A systematic review was to assess the method used in studies examining the impact of heavy metal exposure on pregnancy outcomes. By ensuring a standardized method for study selection and data extraction, the review followed Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2020 guidelines [26], as shown in Figure 1.

2.4 Statistical Models Assessed

The selected studies employed various statistical modeling approaches to assess prenatal heavy metal exposure and birth or early-life outcomes. BKMR, WQS, and SEM were among the methods reported, with their frequencies determined by re-extraction of all methods used in each study [21, 22]. WQS regression was used in three studies to address multicollinearity among correlated exposures and to identify the most influential metals within mixtures [21]. Structural Equation Modeling (SEM), applied in two studies, enabled the exploration of both direct and indirect effects, particularly through mediating variables such as placental weight [24].

Traditional statistical approaches, including linear and logistic regression, were also employed as foundational models, yet they were generally limited in their capacity to account for nonlinearity, multidimensional nature of environmental mixtures, and multiple co-occurring exposures [24, 27]. Each modeling technique was evaluated based on its methodological strengths, interpretability, and suitability for analyzing the complex structure of prenatal exposure to heavy metals.

This study was based entirely on secondary analysis of published literature and did not involve human subjects or identifiable personal data. Therefore, ethical approval was not required.

3 Results

3.1 Bibliometric Findings

A total of 1,564 records were initially retrieved from the Scopus database. Following document type filtering and relevance screening, 125 publications were retained for bibliometric analysis, while 11 studies fulfilled the eligibility criteria for inclusion in the systematic review. After removing reviews and non-modeling studies, 11 original articles were included in the final synthesis. The number of publications has grown significantly in the last decade, especially post-2015, indicating increased scientific attention to environmental exposures during pregnancy. The United States, China, and India were the leading contributors to this field, collectively accounting for 68% of publications. This trend may reflect stronger research capacity and environmental surveillance systems in these countries. However, regions with high environmental risk—particularly low- and middle-income countries—remain underrepresented in modeling studies. The dominant topics included “birth weight,” “prenatal exposure,” and “metal mixtures,” suggesting that most research focused on fetal growth metrics. Importantly, newer keywords such as “Bayesian Kernel Machine Regression” and “Weighted Quantile Sum” reflect a methodological shift toward more complex statistical tools to address the multifactorial nature of environmental exposures. Co-occurrence analysis of author keywords using VOSviewer revealed five thematic clusters (Figure 2).

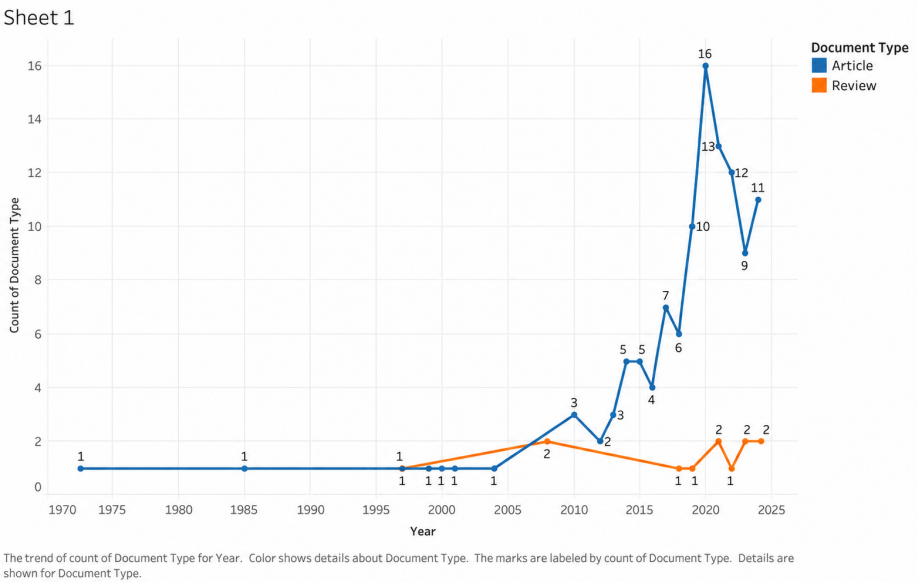


Figure 2. Trends in annual publications concerning the modeling of heavy metal exposure during pregnancy 1973–2023

3.1.1 Geographic distribution of study

The analysis of country contributions showed that the United States accounted for the majority of publications (48 studies), followed by China (40) and Canada (7). Figure 3 shows a country collaboration network. This network shows the global nature of the studies on heavy metal exposure. To aid interpretation of the bibliometric visualizations, the following conventions apply throughout Figures 1–5. In VOSviewer network maps, each node represents a distinct item (e.g., country, author keyword), with node size proportional to its frequency or publication count. Links between nodes indicate co-occurrence or co-authorship relationships; link thickness reflects the strength of association. Colors denote thematic clusters generated by VOSviewer's modularity-based clustering algorithm, where items within the same cluster share stronger conceptual or collaborative ties. Distance between nodes approximates similarity: closely positioned items are more strongly related in the underlying data.

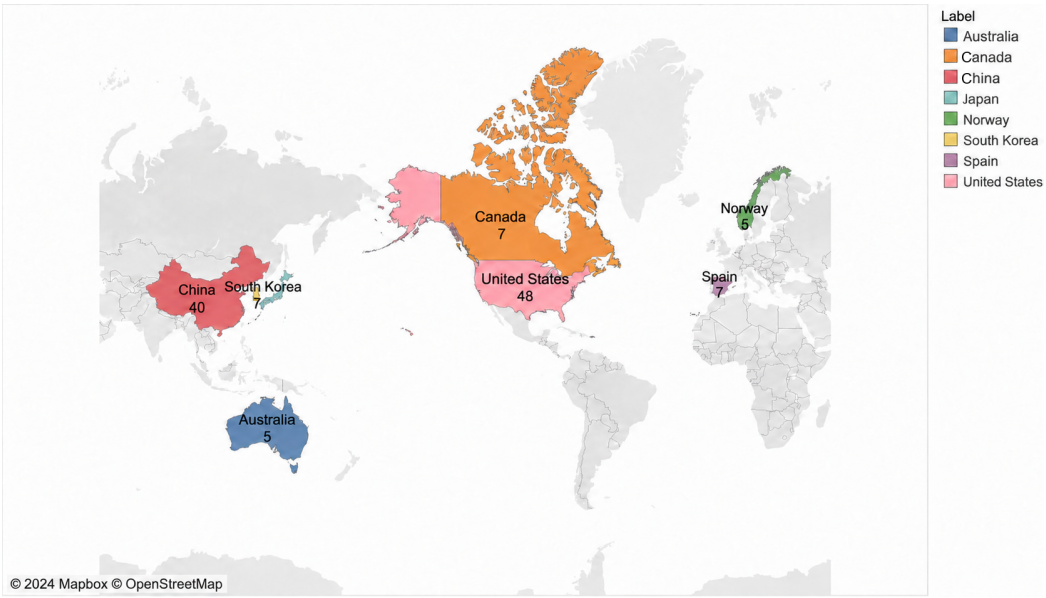


Figure 3. Country distribution map

3.1.2 Keyword co-occurrence analysis

Co-occurrence network analysis of author keywords identified 121 distinct keywords organized into five clusters (Figure 4). The most prominent cluster [green] centered around "pregnancy" and "female" (41 items). In comparison, the red cluster focused on "heavy metal" (43 items) and was connected to studies on "prenatal exposure" and "birth outcomes". This suggests that the primary focus is on the health impacts of heavy metals on pregnancy outcomes, particularly on birth weight and child growth.

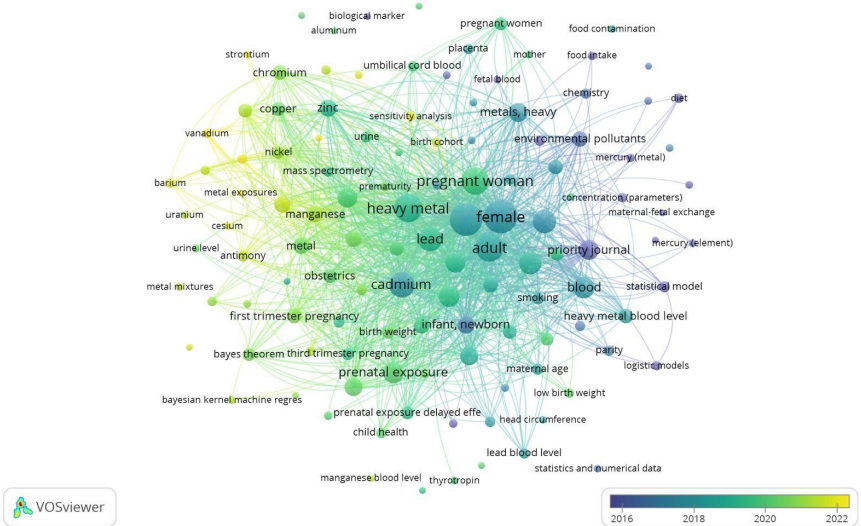


Figure 4. Keyword co-occurrence network generated using VOSviewer

Figure 4 illustrates the keyword co-occurrence network organized into five color-coded clusters. The green cluster (41 items), centered on "pregnancy," "female," and "birth weight," represents the dominant research focus on fetal growth metrics. The red cluster (43 items), linking "heavy metals," "prenatal exposure," and "birth outcomes," reflects the core epidemiological theme of metal toxicity. Notably, the blue/light-green nodes highlight emerging methodological keywords such as "Bayesian Kernel Machine Regression" and "metal mixtures," indicating a shift toward advanced mixture modeling. The purple cluster identifies underexplored metals (e.g., barium, strontium), suggesting opportunities for future investigation.

Figure 5 further illustrates the temporal evolution of these keywords until 2023. The color gradient from blue to yellow indicates the progression of time. This thematic evolution—from single-metal epidemiology to complex mixture analytics—demonstrates the field's methodological maturation and its increasing alignment with real-world exposure scenarios.

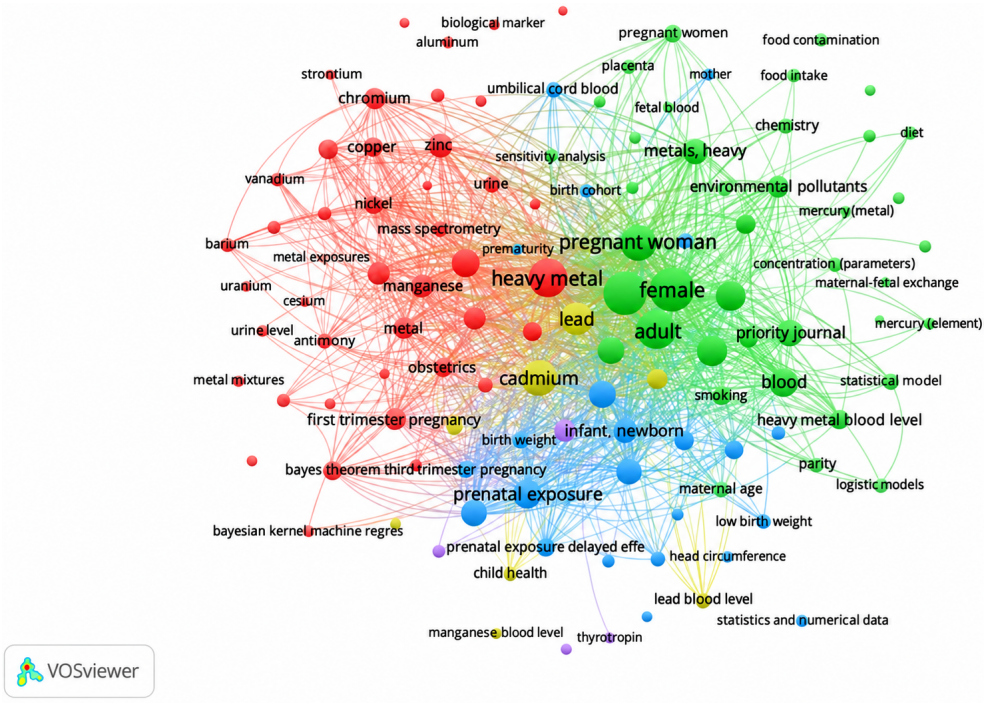


Figure 5. Temporal evolution of author keywords until 2023

The bibliometric findings above provide the context for the systematic review that follows. Specifically, the keyword co-occurrence analysis revealed that "Bayesian Kernel Machine Regression," "Weighted Quantile Sum," and "metal mixtures" have emerged as recent research hotspots, while traditional regression models remain widely used. These observations guided the systematic review to focus on comparing the performance of advanced versus conventional statistical models in assessing prenatal heavy metal exposure. Moreover, the geographic distribution showing underrepresentation of low- and middle-income countries prompted the systematic review to examine whether studies from these regions employed less sophisticated modeling approaches. Thus, the bibliometric analysis not only maps the research landscape but also directly informs the selection of modeling techniques and geographic patterns to be critically evaluated in the systematic review.

3.2 Systematic Review of Modeling Approaches

The bibliometric findings informed the subsequent systematic review by identifying dominant methodological trends, emerging analytical techniques, and geographic disparities requiring further critical evaluation. Eleven studies that employed statistical modeling to assess the effects of prenatal heavy metal exposure were reviewed in-depth. BKMR was the most frequently used model (5/11), particularly in recent studies. BKMR was consistently effective in detecting nonlinear relationships and identifying interactive effects among co-exposed metals, such as cadmium, mercury, and lead. These studies demonstrated that BKMR could predict reduced birth weight and smaller head circumference more accurately than traditional models. WQS regression, used in one study, was beneficial for ranking metals in complex mixtures and identifying dominant contributors to adverse outcomes, including emotional reactivity in children and LBW. However, the model's directional homogeneity assumption limited its interpretability in cases involving both toxic and essential metals. SEM appeared in two studies, allowing for the exploration of

indirect pathways between exposure and outcomes. For example, placental weight was identified as a mediator between cadmium exposure and fetal growth, providing insight into biological mechanisms. Traditional linear and logistic regression models were used across almost all studies either as primary or comparative tools. While they offered baseline associations between single-metal exposure and outcomes such as PTB or SGA, they lacked sensitivity to complex mixture effects and nonlinear dynamics. Table 1 provides a detailed summary of the reviewed studies, including modeling techniques and outcomes.

Table 1. Summary of modeling of heavy metal exposure in pregnant women

References	Country	Name of Journal/ Source (Publisher) and Year of publication	Study Objective	Sample Size/Study Design	Measured Outcome Variables	Model	Key Finding Across Model
Michael et al. [9]	Israel	Environmental Health/ Springer Nature Link/ 2022	To investigate the relationship between prenatal exposure to different types of heavy metals and heavy metal mixtures and anthropometric measurements in children.	975 pairs of mother and children/ cross-sectional	Birth weight, birth head circumference, birth length, Sociodemographics of participants, characteristics of current Pregnancy.	Bayesian Kernel Machine Regression (BKMR)	The negative effects of heavy metals on birth anthropometry were significant, particularly at mixed exposures.
Cowel et al. [28]	American	Environmental Epidemiology/ Wolters Kluwer/2021	To investigate the relationship between urinary metals in late-pregnancy mothers and affective children.	308 mothers/ cohort pairs	Affectivity is a core temperamental trait measured using the Children Behavior Questionnaire-Revised (CBQ-R) and its subscales such as fear, sadness, distress to limitations, and decreased reactivity.	Weighted Quantile Sum (WQS) regression	Metal exposure is associated with increased fear and behavioral inhibition in female children.
Punshon et al. [29]	United State	Environment International/ Science Direct/2019	To analyze arsenic (As), Cd, Hg, and Pb concentrations as part of the New Hampshire Birth Cohort Study, and the potential interaction role of the essential trace elements zinc (Zn) and selenium (Se) as modifiers of these effects. Additionally, the role of placental growth is evaluated and its function as a mediator of placental metal effects on birth weight.	1159 pregnant women/ prospective design	Weight.	Structural Equation Modeling (SEM)	Placental weight mediates the relationship between Cd in the placenta and reduced birth weight.

References	Country	Name of Journal/ Source (Publisher) and Year of publication	Study Objective	Sample Size/Study Design	Measured Outcome Variables	Model	Key Finding Across Model
Zhang et al. [30]	China	Reproductive Toxicology/ Science Direct/2018	To investigate whether Cd exposure during pregnancy is associated with an increased risk of adverse birth outcomes in a sex-dependent manner.	The concentration of Cd in maternal urine samples (U-Cd) was measured in 237 subjects from Guiyu (e-waste area) and 212 subjects from Haojiang/cohort	Birth weight, body length, head circumference.	Linear Regression Model	Prenatal heavy metal exposure decreases children birth weight and length.
Guo et al. [31]	China	Science of the Total Environment/ Science Direct/2016	To evaluate whether Cd exposure in maternal urine and cord blood is associated with birth size parameters.	A birth cohort study including 1073 newborn mother pairs was conducted from 2009 to 2010 at a population farm in China/cohort	Maternal and children exposure to Cd and birth indicators, including birth weight, length, head circumference, and ponderal index.	Generalized Linear Models	Cd in cord blood is associated with a lower ponderal index, particularly in male children.
Wai et al. [32]	Myanmar	International Journal Environmental and Public Health/ MDPI/2017	To identify the potential impact of prenatal exposure to heavy metals on birth outcomes among the Myanmar population.	419 pregnant/ cohort women	Gestational age, Cd exposure, number of maternal births.	Multiple Logistic Regression Models	Prenatal exposure to Cd increases the risk of low birth weight with OR = 1.10.
Inadera et al. [33]	Japan	Environmental Research/ Science Direct/2020	To analyze the relationship between maternal blood Cd levels at different sampling times and sex-dependent children birth size, weight, length, chest circumference, and head, including SGA.	13,969 mother-infant pairs for SGA analysis/cohort	Birth weight, birth length, head circumference, chest circumference, and SGA.	Multiple Linear Regression and Multiple Logistic Regression	Maternal blood Cd levels are associated with low birth weight and increased risk of SGA children.
Ren et al. [34]	China	Environmental International/ Science Direct/2021	To investigate the relationship between hair trace element levels and the risk of PB in pregnant women.	2731 pregnant women/ cases control	Mixture of metals.	BKMR Models and Logistic Regression Analysis	Fe and Zn in hair have a negative association with the risk of PTB, particularly in spontaneous PTB.

References	Country	Name of Journal/ Source (Publisher) and Year of publication	Study Objective	Sample Size/Study Design	Measured Outcome Variables	Model	Key Finding Across Model
Shih et al. [21]	United States	Environmental International/ Science Direct/2021	To evaluate the association of exposure to Cd, Pb, manganese (Mn), Se, and total mercury (Hg) during pregnancy with child growth using data from mother-infant pairs participating in the National Children's Study.	618 mothers pairs/cohort	Gestational age, birth weight, birth length, head circumference, and ponderal index.	Multivariable Linear Regression Model	Hg and Pb are associated with reduced birth weight, while Se is positively associated with birth weight.
Lee et al. [22]	Bangladesh	Environmental Health Perpective/ Science Direct/2021	To assess the relationship between mixed metals (As, Cd, Mn, Pb) in umbilical cord blood and neonate size in Bangladeshi children.	1,088 pregnant women/ cohort	Birth length. Birth head circumference.	Linear Regression Model	Metal mixtures negatively affect birth length and head circumference, particularly Cd and Mn.
Wu et al. [17]	China	Ecotoxicology and Environmental Safety/ Science Direct/2023	To investigate the sex-specific association between prenatal exposure to mixed metals and birth weight in a Chinese birth cohort.	1,296 boys and 1,098 girls/cohort	Concentration of 10 metals As, Cd, Co (cobalt), Cr, Mo (molybdenum), Ni, Pb, Sb (antimony), Se, and Ti (Titanium), with sex, birth weight.	BKMR	Significant negative association of Cd and Tl on BW for GA in female infants.

Note: Sample sizes reflect final analytic cohorts after attrition. Study designs were harmonized per STROBE guidelines. Outcome variables were standardized to WHO/IHME definitions for cross-study comparability.

3.3 Comparison of Statistical Models

To move beyond descriptive summaries, we evaluated each modeling approach across four critical dimensions: sample size requirements, capacity to detect interaction/non-linear effects, suitability for high-dimensional exposure data, and practical implementation constraints. BKMR demonstrates superior performance in high-dimensional settings (typically $\geq$5 correlated metals) and excels at uncovering complex, non-linear dose–response curves and synergistic/antagonistic interactions without prior structural assumptions [21, 22]. Its primary limitation is the assumption of directional homogeneity, which obscures opposing biological effects (e.g., toxic vs. protective metals) and limits its ability to model complex interactions. SEM offers unique advantages for testing predefined causal or mediation pathways but is poorly suited for exploratory interaction detection. It demands strong theoretical justification and large samples relative to the number of estimated parameters (typically $\geq$10–20 observations per parameter), and results are highly sensitive to model misspecification [19, 28]. In contrast, traditional linear/logistic regression remains accessible but systematically underestimates mixture effects due to linearity assumptions and vulnerability to multicollinearity [24, 29, 30]. Table 2 has been updated to reflect these concrete operational distinctions, providing researchers with actionable guidance for method selection based on study design, data structure, and computational capacity.

Table 2. Comparative evaluation of statistical models

Model	Optimal Application Conditions	Practical Difficulties & Constraints
BKMR	Complex mixtures with suspected non-linear or interactive effects	High computational demand; MCMC convergence sensitivity; steep learning curve for interpretation
WQS regression	Ranking key contributors in highly correlated exposure datasets	Assumes directional homogeneity; masks opposing toxic/protective effects; requires $\geq$100 bootstrap iterations for stable weights
SEM	Testing predefined causal/mediation pathways (e.g., placental, epigenetic)	Requires strong theoretical framework & large N; highly sensitive to misspecification; needs validated measurement models
Linear/logistic regression	Single-exposure screening, baseline comparisons, or resource-limited settings	Fails with multicollinearity; assumes linearity/additivity; underestimates combined mixture risks

Note: BKMR = Bayesian Kernel Machine Regression; WQS = Weighted Quantile Sum; SEM = Structural Equation Modeling; MCMC = Markov chain Monte Carlo.

4 Discussion

This study presents a comprehensive synthesis of global research trends and modeling approaches used to evaluate prenatal heavy metal exposure and its effects on pregnancy outcomes. Through the integration of bibliometric analysis and systematic review, this research not only maps the scholarly landscape but also identifies key methodological advancements, emerging gaps, and opportunities for future investigation.

Our bibliometric findings indicate a significant increase in publications since 2015, with the United States, China, and India emerging as leading contributors. This pattern reflects global recognition of the health risks associated with environmental pollutants and the critical need to improve maternal and child health indicators. However, the geographical imbalance suggests a disparity in environmental monitoring capacity and research funding, particularly in low- and middle-income countries (LMICs), which often bear a disproportionate burden of heavy metal exposure [10, 16]. This underrepresentation likely stems from limited research infrastructure, insufficient funding for environmental epidemiology, and the absence of routine biomonitoring programs in many LMICs, despite their often higher burden of heavy metal exposure from uncontrolled industrial and agricultural activities. Increased international collaboration and capacity-building efforts in LMICs are essential to ensure equitable advancements in environmental epidemiology.

Our systematic evaluation confirms that the methodological shift toward advanced mixture models reflects a critical need to move beyond single-exposure paradigms. The consistent identification of synergistic Cd–Pb–Hg interactions across BKMR applications underscores that traditional linearity assumptions systematically underestimate real-world toxicity thresholds [21–23]. This has direct implications for environmental risk assessment, where regulatory guidelines remain predominantly metal-specific rather than mixture-based [21, 24]. WQS regression successfully highlighted dominant contributors in correlated exposure profiles, yet its reliance on directional homogeneity proved limiting when essential and toxic metals co-occurred a frequent scenario in LMIC populations with mixed nutritional and contaminant burdens [21, 28, 29]. SEM uniquely bridged statistical outputs with biological plausibility by quantifying placental mediation, demonstrating how advanced frameworks can elucidate mechanistic pathways beyond mere association [24, 29]. Conversely, the persistent reliance on traditional regression models in recent cohorts highlights a methodological inertia that likely obscures complex exposure–response dynamics. Collectively, these findings emphasize that model selection must be driven by exposure architecture, sample characteristics, and the specific public health question rather than analytical convenience [30].

Beyond differences in statistical models, the reviewed studies varied considerably by study location, exposure levels, and population characteristics, which may influence observed associations. Studies in highly polluted areas such as the e-waste site in Guiyu, China [30] and Jepara coastal area of Indonesia [16] reported higher maternal metal burdens and stronger negative effects on birth weight than studies in less contaminated settings, such as Japan [18] or the United States [29]. Population characteristics also differed [19], low maternal BMI and food insecurity likely exacerbated cadmium's impact on LBW (OR = 1.10), whereas in the well-nourished New Hampshire cohort [29], placental mediation partially offset metal toxicity. Sex-specific effects were also notable: studies from China [17, 31] found stronger cadmium associations with reduced birth size in male infants, while Cowel et al. [28] reported that metal mixtures increased fear-related behaviors predominantly in female children. These contextual differences underscore the need to interpret modeling results within specific environmental and demographic settings and to stratify future analyses by sex, nutrition, and geographic region.

Consistent with previous meta-analyses, this review reaffirms the association between cadmium exposure and

LBW, as well as between mercury exposure and impaired neurodevelopment [11, 17, 18, 31, 35–37]. Arsenic exposure was notably linked to spontaneous miscarriage, and lead was identified as a primary contributor to congenital anomalies through additive and interactive effects [23, 38, 39]. These findings are significant in the context of Sustainable Development Goal 3, which emphasizes the reduction of maternal and neonatal mortality and morbidity linked to hazardous chemical exposure [25, 35, 36].

Previous reviews have examined the associations between prenatal mercury and arsenic exposure and fetal growth and pregnancy outcomes [37, 38]. For example, cadmium impairs placental nutrient transport, while lead and mercury interfere with neurodevelopmental gene expression and hormonal regulation. These mechanisms underscore the need for integrated biomonitoring frameworks that combine exposure assessment with biological markers of effect [39–41].

Despite methodological advancements, several challenges persist in the current literature. Many studies lack rigorous sensitivity analyses, fail to adjust for critical confounders (e.g., maternal nutrition, socioeconomic status), or do not incorporate time-varying exposures. Few models address sex-specific effects or stratify risks by critical windows of susceptibility during gestation, such as the first trimester or organogenesis period [42]. These limitations hinder causal inference and reduce the utility of findings for risk assessment and policy translation.

Future research should prioritize longitudinal cohort designs with repeated exposure measures, integration of omics data (e.g., metabolomics, epigenomics), and use of ensemble machine learning methods that can enhance model prediction and reduce overfitting. Moreover, collaboration across disciplines environmental science, epidemiology, toxicology, and biostatistics—will be essential to develop robust, scalable models capable of informing regulatory frameworks.

This review has several limitations. First, only 11 studies were included, which may limit the statistical power and generalizability of the findings. Second, most studies were conducted in high-income countries, which may limit the applicability of the findings to low- and middle-income settings. Therefore, the findings regarding the performance of BKMR and WQS compared with traditional regression should be interpreted with caution. Further studies involving larger and more diverse populations are needed to confirm these findings.

The growing application of advanced statistical approaches such as BKMR, WQS regression, and SEM has important implications for environmental health risk assessment. Unlike conventional regression models, these approaches are capable of evaluating complex exposure mixtures, nonlinear dose–response relationships, and interaction effects among multiple heavy metals. Such capabilities are particularly relevant in real-world environmental settings where pregnant women are simultaneously exposed to various contaminants rather than single substances in isolation. Improved identification of cumulative and interactive exposure effects may support more accurate risk characterization, early identification of vulnerable populations, and evidence-based environmental health interventions. Furthermore, these methodological advances may contribute to the development of more comprehensive maternal and child health protection policies, particularly in regions with high environmental contamination burdens.

In conclusion, this review highlights the growing application and value of advanced statistical models in understanding the impacts of mixed heavy metal exposures on pregnancy outcomes. Models such as BKMR and WQS represent a paradigm shift in environmental health research, offering tools that better reflect real-world exposure conditions and biological complexity. However, standardization of modeling practices, expansion of research in underrepresented regions, and incorporation of mechanistic endpoints remain urgent priorities to fully harness the potential of these analytic innovations.

5 Conclusions

This study underscores a significant methodological shift from traditional regression techniques to advanced mixture-based models, such as BKMR and WQS regression, in the investigation of prenatal heavy metal exposure. These advanced modeling approaches can facilitate the investigation of complex exposure–outcome relationships, including potential nonlinear or interactive associations among metal mixtures, which are challenging to identify using conventional statistical methods, but the evidence remains heterogeneous and does not establish superior predictive performance over conventional approaches. The findings also highlight notable geographic disparities in research productivity and methodological application, with limited representation from low- and middle-income countries, despite their high exposure burdens. Future research should prioritize longitudinal designs, repeated biomonitoring assessments, and the broader adoption of advanced modeling approaches to enhance environmental risk assessment and support evidence-based maternal health policies.

Author Contributions

Conceptualization, H.A.; data curation, A.U.Z. and H.A.; formal analysis, A.U.Z., H.A., A.B.B., S.S., R.P.A., and A.K.; funding acquisition, A.U.Z., H.A., and A.B.B.; investigation, A.U.Z., H.A., and A.B.B.; methodology, H.A., A.B.B., S.S., R.P.A., and A.K.; project administration, A.U.Z., H.A., and A.B.B.; resources, H.A.; software,

A.U.Z., H.A., and A.B.B.; supervision, H.A., A.B.B., S.S., R.P.A., and A.K.; validation, H.A.; visualization, A.U.Z. and A.B.B.; writing—original draft preparation, A.U.Z. and I.S.; writing—review and editing, A.U.Z., H.A., A.B.B., and I.S. All authors have read and agreed to the published version of the manuscript.

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Data Availability

Not applicable.

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Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] S. Iliodromiti, D. F. Mackay, G. C. S. Smith, J. P. Pell, N. Sattar, D. A. Lawlor, and S. M. Nelson, "Customised and noncustomised birth weight centiles and prediction of stillbirth and infant mortality and morbidity: A cohort study of 979,912 term singleton pregnancies in Scotland," *PLoS Med.*, vol. 14, no. 1, pp. 1–16, 2017. https://doi.org/10.1371/journal.pmed.1002228

[2] C. N. Amadi, Z. N. Igweze, and O. E. Orisakwe, "Heavy metals in miscarriages and stillbirths in developing nations," *Middle East Fertil. Soc. J.*, vol. 22, no. 2, pp. 91–100, 2017. https://doi.org/10.1016/j.mefs.2017.03.003

[3] D. Kabiri, R. Romero, D. W. Gudicha, E. H. Andrade, P. Pacora, N. Benshalom-Tirosh, D. Tirosh, and L. Yeo, "Prediction of adverse perinatal outcome by fetal biometry: Comparison of customized and population-based standards," *Ultrasound Obstet. Gynecol.*, vol. 55, no. 2, pp. 177–188, 2020. https://doi.org/10.1002/uog.20299

[4] M. H. Forouzanfar, A. Afshin, L. T. Alexander, H. R. Anderson, Z. A. Bhutta, S. Biryukov, M. Brauer, R. Burnett, K. Cercy, F. J. Charlson *et al.*, "Global, regional, and national comparative risk assessment of 79 behavioural, environmental and occupational, and metabolic risks or clusters of risks, 1990–2015: A systematic analysis for the Global Burden of Disease Study 2015," *Lancet*, vol. 388, no. 10053, pp. 1659–1724, 2016. https://doi.org/10.1016/S0140-6736(16)31679-8

[5] I. Manisalidis, E. Stavropoulou, A. Stavropoulos, and E. Bezirtzoglou, "Environmental and health impacts of air pollution: A review," *Front. Public Health*, vol. 8, no. 14, 2020. https://doi.org/10.3389/fpubh.2020.00014

[6] C. Liu, P. Yin, R. Chen, X. Meng, L. Wang, Y. Niu, Z. Lin, Y. Liu, J. Liu, J. Qi *et al.*, "Ambient carbon monoxide and cardiovascular mortality: A nationwide time-series analysis in 272 cities in China," *Lancet Planet. Health*, vol. 2, no. 1, pp. e2–e3, 2018. https://doi.org/10.1016/S2542-5196(17)30181-X

[7] M. E. Bianchi and J. M. Restrepo, "Low birthweight as a risk factor for non-communicable diseases in adults," *Front. Med.*, vol. 8, pp. 6–11, 2022. https://doi.org/10.3389/fmed.2021.793990

[8] A. L. Hinwood, A. C. Callan, M. Ramalingam, M. Boyce, J. Heyworth, P. McCafferty, and J. Ø. Odland, "Cadmium, lead and mercury exposure in non smoking pregnant women," *Environ. Res.*, vol. 126, pp. 118–124, 2013. https://doi.org/10.1016/j.envres.2013.07.005

[9] T. Michael, E. Kohn, S. Daniel, A. Hazan, M. Berkovitch, A. Brik, O. Hochwald, L. Borenstein-Levin, M. Betser, M. Moskovich *et al.*, "Prenatal exposure to heavy metal mixtures and anthropometric birth outcomes: A cross-sectional study," *Environ. Health*, vol. 21, no. 1, p. 139, 2022. https://doi.org/10.1186/s12940-022-00950-z

[10] WHO, "Exposure to cadmium: A major public health concern," World Health Organization, 2019. https://www.who.int/publications/i/item/WHO-CED-PHE-EPE-19-4-3

[11] S. S. Zinia, K. H. Yang, E. J. Lee, M. N. Lim, J. Kim, and W. J. Kim, "Effects of heavy metal exposure during pregnancy on birth outcomes," *Sci. Rep.*, vol. 13, no. 1, pp. 1–10, 2023. https://doi.org/10.1038/s41598-023-46271-0

[12] C. Z. Yuan and X. R. Wang, "Source apportionment and health risk assessment of heavy metals in soils of old industrial areas: A case study of Shanghai, China," *Int. J. Environ. Res. Public Health*, vol. 20, no. 3, 2023. https://doi.org/10.3390/ijerph20032395

[13] A. Altıkulac and S. Turhan, "Evaluation of some potentially toxic elements and associated ecological and health risks in topsoil samples adjacent to an industrial zone in Turkey," *ACS Omega*, 2024. https://doi.org/10.1021/acsomega.3c07638

[14] I. Irawati, H. Amqam, R. Indriasari, A. B. Birawida, M. Masni, S. Werorilangi, and I. Suryadi, "Heavy metal exposure in pregnancy and the impact on fetal development: Five decades of global research through bibliometric analysis," *Int. J. Environ. Impacts*, vol. 9, no. 1, pp. 14–26, 2026. https://doi.org/10.56578/ijei090102

[15] T. Sun, Z. Zheng, M. Yang, M. Pan, Q. Tan, Y. Ma, Y. Zhou, M. He, and Y. Sun, "Heavy metal exposure during pregnancy and its association with adverse birth outcomes: A cross-sectional study," *Geohealth*, vol. 9, no. 10, 2025. https://doi.org/10.1029/2025GH001471

[16] M. Muniroh, A. Nugraheni, M. Mulyono, Y. Nindita, F. Swastawati, S. Bakri, A. R. Gumay, H. Amqam, E. D. Nugraha, A. C. N. Marchianti *et al.*, "Estimating daily intake of heavy metals from fish consumption and its association with hemoglobin levels in Indonesian pregnant women," *Discov. Appl. Sci.*, vol. 8, p. 233, 2026. https://doi.org/10.1007/s42452-026-08239-3

[17] Y. Wu, F. Zeng, J. Li, Y. Jiang, S. Zhao, L. D. Knibbs, X. Zhang, Y. Wang, Q. Zhang, Q. Wang *et al.*, "Sex-specific relationships between prenatal exposure to metal mixtures and birth weight in a Chinese birth cohort," *Ecotoxicol. Environ. Saf.*, vol. 262, p. 115158, 2023. https://doi.org/10.1016/j.ecoenv.2023.115158

[18] T. Takatani, A. Eguchi, M. Yamamoto, K. Sakurai, R. Takatani, Y. Taniguchi, S. F. Nakayama, C. Mori, M. Kamijima *et al.*, "Individual and mixed metal maternal blood concentrations in relation to birth size: An analysis of the Japan Environment and Children's Study (JECS)," *Environ. Int.*, vol. 165, 2022. https://doi.org/10.1016/j.envint.2022.107318

[19] J. Grzesik-Gąsior, J. Sawicki, A. Pieczykolan, and A. Bień, "Content of selected heavy metals in the umbilical cord blood and anthropometric data of mothers and newborns in Poland: Preliminary data," *Sci. Rep.*, vol. 13, no. 1, pp. 1–10, 2023. https://doi.org/10.1038/s41598-023-41249-4

[20] J. Liu, F. Ruan, S. Cao, Y. Li, S. Xu, and W. Xia, "Associations between prenatal multiple metal exposure and preterm birth: Comparison of four statistical models," *Chemosphere*, vol. 289, p. 133015, 2022. https://doi.org/10.1016/j.chemosphere.2021.133015

[21] Y. H. Shih, H. Y. Chen, K. Christensen, A. Handler, M. E. Turyk, and M. Argos, "Prenatal exposure to multiple metals and birth outcomes: An observational study within the National Children's Study cohort," *Environ. Int.*, vol. 147, p. 106373, 2021. https://doi.org/10.1016/j.envint.2020.106373

[22] M. S. Lee, K. D. Eum, G. Mostofa, Q. Quamruzzaman, M. L. Kile, M. Mazumdar, and D. C. Christiani, "Umbilical cord blood metal mixtures and birth size in Bangladeshi children," *Environ. Health Perspect.*, vol. 129, no. 5, 2021. https://doi.org/10.1289/EHP7502

[23] Z. Nie, H. Xu, M. Qiu, M. Liu, C. Chu, M. S. Bloom, and Y. Ou, "Associations of maternal exposure to multiple plasma trace elements with the prevalence of fetal congenital heart defects: A nested case-control study," *Sci. Total Environ.*, vol. 912, p. 169409, 2024. https://doi.org/10.1016/j.scitotenv.2023.169409

[24] J. Grzesik-Gąsior, A. Bień, K. Zalewska, M. Nieszporek, K. Witkowska, and A. Merklinger-Gruchała, "Maternal–fetal exposure to oncoelements and their oxidative and epigenetic impact on pregnancy outcomes," *Int. J. Mol. Sci.*, vol. 27, no. 2, p. 669, 2026. https://doi.org/10.3390/ijms27020669

[25] X. Sun, W. Liu, B. Zhang, X. Shen, C. Hu, X. Chen, S. Jin, Y. Jiang, H. Liu, Z. Cao *et al.*, "Maternal heavy metal exposure, thyroid hormones, and birth outcomes: A prospective cohort study," *J. Clin. Endocrinol. Metab.*, vol. 104, no. 11, pp. 5043–5052, 2019. https://doi.org/10.1210/jc.2018-02492

[26] M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, T. C. Hoffmann, C. D. Mulrow, L. Shamseer, J. M. Tetzlaff, E. A. Akl, S. E. Brennan *et al.*, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *Int. J. Surg.*, vol. 88, p. 105906, 2021. https://doi.org/10.1016/j.ijsu.2021.105906

[27] S. Sabra, E. Malmqvist, A. Saborit, E. Gratacós, and M. D. Gomez Roig, "Heavy metals exposure levels and their correlation with different clinical forms of fetal growth restriction," *PLoS ONE*, vol. 12, no. 10, p. e0185645, 2017. https://doi.org/10.1371/journal.pone.0185645

[28] W. Cowell, E. Colicino, Y. Levin-Schwartz, M. Bosquet Enlow, C. Amarasiriwardena, S. S. Andra, C. Gennings, R. O. Wright, and R. J. Wright, "Prenatal metal mixtures and sex-specific infant negative affectivity," *Environ. Epidemiol.*, vol. 5, no. 2, 2021. https://doi.org/10.1097/EE9.0000000000000147

[29] T. Punshon, Z. Li, B. P. Jackson, W. T. Parks, M. Romano, D. Conway, E. R. Baker, and M. R. Karagas, "Placental metal concentrations in relation to placental growth, efficiency and birth weight," *Environ. Int.*, vol. 126, pp. 533–542, 2019. https://doi.org/10.1016/j.envint.2019.01.063

[30] Y. Zhang, X. Xu, A. Chen, C. B. Davuljigari, X. Zheng, S. S. Kim, K. N. Dietrich, S. M. Ho, T. Reponen, and X. Huo, "Maternal urinary cadmium levels during pregnancy associated with risk of sex-dependent birth outcomes from an e-waste pollution site in China," *Reprod. Toxicol.*, vol. 75, pp. 49–55, 2018. https:

//doi.org/10.1016/j.reprotox.2017.11.003

[31] J. Guo, C. Wu, X. Qi, S. Jiang, Q. Liu, J. Zhang, Y. Cao, X. Chang, and Z. Zhou, "Adverse associations between maternal and neonatal cadmium exposure and birth outcomes," *Sci. Total Environ.*, vol. 575, pp. 581–587, 2017. https://doi.org/10.1016/j.scitotenv.2016.09.016

[32] K. M. Wai, O. Mar, S. Kosaka, M. Umemura, and C. Watanabe, "Prenatal heavy metal exposure and adverse birth outcomes in Myanmar: A birth-cohort study," *Int. J. Environ. Res. Public Health*, vol. 14, p. 1339, 2017. https://doi.org/10.3390/ijerph14111339

[33] H. Inadera, A. Takamori, K. Matsumura, A. Tsuchida, Z.-G. Cui, K. Hamazaki, T. Tanaka, M. Ito, M. Kigawa, H. Origasa *et al.*, "Association of blood cadmium levels in pregnant women with infant birth size and small for gestational age infants: The Japan Environment and Children's study," *Environ. Res.*, vol. 191, p. 110007, 2020. https://doi.org/10.1016/j.envres.2020.110007

[34] M. Ren, J. Zhao, B. Wang, H. An, Y. Li, X. Jia, J. Wang, S. Wang, L. Yan, X. Liu *et al.*, "Associations between hair levels of trace elements and the risk of preterm birth among pregnant women: A prospective nested case-control study in Beijing Birth Cohort (BBC), China," *Environ. Int.*, vol. 158, p. 106965, 2022. https://doi.org/10.1016/j.envint.2021.106965

[35] A. K. Amegah, C. Sewor, and J. J. K. Jaakkola, "Cadmium exposure and risk of adverse pregnancy and birth outcomes: A systematic review and dose-response meta-analysis of cohort and cohort-based case-control studies," *J. Expo. Sci. Environ. Epidemiol.*, vol. 31, no. 2, pp. 299–317, 2021. https://doi.org/10.1038/s41370-021-00289-6

[36] M. Khoshhali, N. Rafiei, Z. Farajzadegan, B. Shoshari-Yeganeb, and R. Kelishadi, "Maternal exposure to cadmium and fetal growth: A systematic review and meta-analysis," *Biol. Trace Elem. Res.*, vol. 195, no. 1, pp. 9–19, 2020. https://doi.org/10.1007/s12011-019-01819-y

[37] K. Dack, M. Fell, C. M. Taylor, A. Havdahl, and S. J. Lewis, "Mercury and prenatal growth: A systematic review," *Int. J. Environ. Res. Public Health*, vol. 18, no. 13, 2021. https://doi.org/10.3390/ijerph18137140

[38] R. Quansah, F. A. Armah, D. K. Essumang, I. Luginaah, E. Clarke, K. Marfoh, S. J. Cobbina, E. Nketiah-Amponsah, P. B. Namujju, S. Obiri *et al.*, "Association of arsenic with adverse pregnancy outcomes/infant mortality: A systematic review and meta-analysis," *Environ. Health Perspect.*, vol. 123, no. 5, pp. 412–421, 2015. https://doi.org/10.1289/ehp.1307894

[39] Z. Wang, S. Huang, W. Zhang, X. Zeng, C. Chu, Q. Li, X. Cui, Q. Wu, G. Dong, J. Huang *et al.*, "Chemical element concentrations in cord whole blood and the risk of preterm birth for pregnant women in Guangdong, China," *Ecotoxicol. Environ. Saf.*, vol. 247, p. 114228, 2022. https://doi.org/10.1016/j.ecoenv.2022.114228

[40] S. S. Kim, X. Xu, Y. Zhang, X. Zheng, R. Liu, K. N. Dietrich, T. Reponen, C. Xie, H. Sucharew, X. Huo *et al.*, "Birth outcomes associated with maternal exposure to metals from informal electronic waste recycling in Guiyu, China," *Environ. Int.*, vol. 137, p. 105580, 2020. https://doi.org/10.1016/j.envint.2020.105580

[41] L. A. Eaves, C. M. Bulka, J. E. Rager, A. J. Gardner, A. L. Galusha, P. J. Parsons, T. M. O'Shea, and R. C. Fry, "Metal mixtures modeling identifies birth weight-associated gene networks in the placentas of children born extremely preterm," *Chemosphere*, vol. 313, pp. 1–28, 2023. https://doi.org/10.1016/j.chemosphere.2022.137469

[42] J. F. Bobb, B. Claus Henn, L. Valeri, and B. A. Coull, "Statistical software for analyzing the health effects of multiple concurrent exposures via Bayesian kernel machine regression," *Environ. Health*, vol. 17, no. 1, pp. 1–10, 2018. https://doi.org/10.1186/s12940-018-0413-y